

The P6 (Port 6) Truss, launched in November 2000, brought the first huge solar arrays to the ISS. Over the next decade, more trusses would be added that increased functionality of the ISS. The SO (Starboard O) Truss acts as the junction from which external utilities – such as power, data, video and ammonia for the Active Thermal Control System – are routed to the pressurized modules. The S1 Truss and the P1 Truss provide structural support for the Active Thermal Control System, Mobile Transporter and one Crew and Equipment Translation Aid cart. The other trusses are used to attach equipment as well as additional solar arrays for enhanced electricity generation, storage and transfer.

Destiny

In February 2001, the US-made Destiny module was attached to the ISS, making it the primary research laboratory for US payloads, supporting a wide range of experiments and studies contributing to health, safety and quality of life for people all over the world. Destiny also featured a large window – an optical gem that makes it possible to shoot very high quality photos and video. Station crewmembers use video and still cameras at the window to record Earth's ever-changing landscapes below. In addition to its role as a science facility, Destiny also contains the control centre for the Station's robotic arm operations.

Canadarm 2 –

Canada's most major contribution to the ISS is a modified version of the robotic arm used on the space shuttles, launched in April 2001. Canadarm 2 is a robotic system that plays a key role in space station assembly and maintenance: moving equipment and supplies around the station, supporting astronauts working in space, and servicing instruments and other payloads attached to the space station. Astronauts receive robotics training to enable them to perform these functions with the arm. It is 17.6 metres long when fully extended and has seven motorised joints. This arm is capable of handling large payloads and assisting with docking the space shuttle.

Quest

The Quest Joint Airlock is a pressurised flight element consisting of two cylindrical chambers attached end-to-end by a connecting bulkhead and hatch. The airlock is the primary path for ISS spacewalk entry and departure. The addition of the airlock permits Space Station-based spacewalks to be performed without major loss of environmental consumables such as air. Quest was attached to the ISS on